Predictive Customer Churn Analytics and Retention Optimization in E-Commerce Using Machine Learning.

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My GitHub Repository: sba26137-ux/strategic-thinking-capstone-project-proposal

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1. Introduction

In today's e-commerce industry, competition is fierce, and customer loyalty is increasingly difficult to maintain. Customers can easily switch to a competitor with just a few clicks. Because of this, customer retention has become a major strategic focus for online retailers. Bringing in new customers requires significant marketing spend on advertising and promotions (Gupta et al., 2004). Research shows that acquiring a new customer is roughly five times more expensive than keeping an existing one, while a small 5% increase in customer retention can noticeably improve overall profitability (Reichheld, 2001).

This Capstone project addresses the problem of customer churn in e-commerce. Churn happens when an active customer stops buying from a platform or moves to a competitor. Instead of waiting for customers to leave, this project uses data analytics and machine learning to identify warning signs early on. By analyzing past transactions, customer satisfaction ratings, and platform activity, the goal is to build a model that predicts which customers are at risk of leaving, allowing marketing teams to take targeted action before it is too late.

2. Problem Definition

Customer churn creates a direct financial problem for online stores. When customers leave, companies lose future sales, lower their Customer Lifetime Value (CLV), and waste the money spent to acquire those customers in the first place.

Most e-commerce retention efforts are reactive. Companies usually send discount codes or win-back emails only after a customer has been inactive for months or has closed their account. At that point, it is often too late to change their mind, making marketing efforts ineffective and costly.

The main analytical challenge is that static reports and basic spreadsheets cannot easily catch subtle changes in customer behavior. E-commerce platforms collect large amounts of data—such as login history, purchase frequency, product reviews, and customer service complaints—but this data is rarely used to predict future actions.

By applying supervised machine learning, this project solves that problem. It converts raw customer interaction data into clear risk scores, helping managers focus their retention budgets on high-risk, high-value customers who are actually likely to stay if given the right incentive.

3. Aim and Objectives

The main goal of this two-semester project is to build and evaluate a machine learning model that predicts customer churn in e-commerce, and to integrate these predictions into an interactive Power BI dashboard for strategic decision-making.

Specific Objectives

1. **Literature & Strategy Review:** Review academic literature and practical business frameworks on customer churn, retention metrics, and the use of predictive analytics in business decision-making.
2. **Data Cleaning & Feature Engineering:** Preprocess the e-commerce dataset by handling missing values, removing outliers, encoding variables, and creating new features (e.g., time since last purchase, complaint ratios).
3. **Exploratory Data Analysis (EDA) & SQL Queries:** Analyze customer patterns using Python and SQL to identify key differences in purchasing habits between churned and retained customers.
4. **Predictive Model Development:** Train and compare several classification models (Logistic Regression, Decision Trees, Random Forest, and XGBoost) using metrics like Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
5. **Interactive BI Dashboard Creation:** Build a Power BI dashboard that presents the model's outputs through clear KPIs, risk categories, and visual reports for marketing managers.
6. **Business Recommendations:** Translate model insights into actionable retention strategies based on feature importance rankings.

Business Hypotheses

To test the findings statistically, the project sets out the following hypotheses:

- **Hypothesis 1 (H_0):** Customer service complaints, order frequency, and satisfaction scores have no significant effect on customer churn.
- **Hypothesis 1 (H_1):** Customers with logged complaints, declining order frequency, and low satisfaction scores have a significantly higher probability of churning ($p < 0.05$).

4. Scope

To keep the project realistic and manageable over two semesters, clear boundaries have been set.

In-Scope

- **Data Processing:** Cleaning, transforming, and analyzing a secondary e-commerce customer dataset using Python and SQL.
- **Feature Engineering:** Creating useful indicators from transactional logs, user satisfaction scores, and account details.
- **Machine Learning:** Building, tuning, and evaluating binary classification models (Churn = 1, Retained = 0).
- **Dashboard Deployment:** Developing an interactive Power BI report showcasing churn risk scores, customer segments, and strategic KPIs.
- **Business Evaluation:** Assessing how predictive analytics can improve customer retention strategies and lower acquisition costs.

Out-of-Scope

- **Live System Integration:** Connecting the model directly to live production databases or APIs (e.g., Shopify/Magento) in real time.
- **Automated Marketing Execution:** Setting up automated email or SMS campaign triggers to end customers.
- **Text Mining / NLP:** Analyzing raw, unstructured text from customer service chats or reviews (the project focuses exclusively on structured tabular data).

5. Methodology and System Architecture

Process Methodology

This project follows the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** framework, which breaks data projects down into six logical phases:

1. **Business Understanding:** Identifying business goals and churn metrics.
2. **Data Understanding:** Exploring dataset features and data quality.
3. **Data Preparation:** Cleaning data, encoding categories, and engineering features.
4. **Modelling:** Training classification algorithms on historical data.
5. **Evaluation:** Assessing models using technical metrics and business logic.
6. **Deployment:** Presenting findings and risk models in an interactive dashboard.

Tools and Technologies

- **Python Stack:** Python 3.x, using pandas and numpy for data manipulation, scikit-learn and xgboost for model training, and seaborn / matplotlib for plots.
- **SQL / MySQL Workbench:** For querying, structuring, and exploring tabular data.
- **Power BI:** For building an interactive executive dashboard.
- **GitHub:** For version control, code tracking, and assignment submission.

6. Data Sources

- **Dataset Name:** E-Commerce Customer Churn Analysis and Prediction Dataset (sourced from Kaggle).
- **Dataset Size: 5,630 rows** and **20 columns**, providing a solid dataset size for machine learning without being overly heavy to process.
- **Key Columns:**
    - *Target:* Churn (1 = Churned, 0 = Retained).
    - *Account & Demographics:* CityTier, Gender, MaritalStatus, PreferredLoginDevice, PreferedOrderCat.
    - *Behavioral Features:* Tenure (months with platform), WarehouseToHome (distance), SatisfactionScore (1-5), Complain (0/1 flag), OrderAmountHikeFromlastYear, CouponUsed, OrderCount, DaysSinceLastOrder.
- **Licensing and Permissions:** The dataset is available on Kaggle under the **CC0: Public Domain** license. This allows free academic and educational use. It contains no personally identifiable information (PII).

7. Expected Deliverables

By the end of Semester 2, the project will yield the following deliverables:

1. **GitHub Repository:** A organized GitHub repository containing well-documented Jupyter Notebooks (.ipynb) and Python scripts (.py).
2. **Processed Data & SQL Scripts:** Cleaned datasets along with SQL files used for data preparation and exploratory queries.
3. **Machine Learning Models:** Trained classification models saved as .pkl files, ready to generate churn probabilities on new data.
4. **Interactive Power BI Dashboard:** A .pbix file featuring churn risk breakdowns, customer segments, feature importance highlights, and key KPIs.

5. **Final Capstone Report:** A detailed written project report summarizing the methodology, model performance, business findings, and strategic retention recommendations.

8. Project Timeline

The work is split across two semesters as outlined below:

Phase	Main Activities & Milestones	Target Completion
Semester 1: Research & Preprocessing	• **Milestone 1:** Capstone Project Proposal Submission	Weeks 1–4
	• Literature Review on E-Commerce Analytics & Retention	Weeks 5–8
	• Data Ingestion, Cleaning & SQL Database Setup	Weeks 9–11
	• **Milestone 2:** Exploratory Data Analysis & Feature Engineering	Weeks 12–14
Semester 2: Modelling & Deployment	• Machine Learning Model Development & Optimization	Weeks 1–4
	• Model Evaluation (ROC-AUC, Precision, Recall)	Weeks 5–8
		Weeks 9–11

Phase	Main Activities & Milestones	Target Completion
	• **Milestone 3:** Power BI Dashboard Integration • Final Thesis Writing, Code Cleanup & GitHub Submission	Weeks 12–14

9. Ethical and Legal Considerations

- **Data Privacy & GDPR:** The project uses anonymized secondary data. It complies with GDPR guidelines as it contains no names, email addresses, phone numbers, addresses, or payment details.
- **Algorithmic Fairness:** Predictive models will be checked to avoid unwanted bias against specific customer groups (such as marital status or gender) in retention strategies.
- **Security & Storage:** Project code and data files will be saved securely on password-protected cloud storage and a private/shared GitHub repo following CCT guidelines.
- **Academic Integrity:** All academic papers, books, software libraries, and datasets used will be credited using the **Harvard Referencing System**.

References

- Gupta, S., Lehmann, D.R. and Stuart, J.A., 2004. Valuing customers. *Journal of Marketing Research*, 41(1), pp.7-18.
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- Verbeke, W., Dejaeger, K., Martens, D., Hur, J. and Baesens, B., 2012. New insights into churn prediction in the telecommunication sector: A profit-driven data mining approach. *European Journal of Operational Research*, 218(1), pp.211-229.